
ON NUMBER AND MAGNITUDE

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CONTENTS

CHAPTER 1

COMMENSURABILITY AND THE DIAGONAL PAGE 3

1.1	Magnitudes and Their Measures	3
1.2	The Pythagorean Theorem	3
1.3	The Incommensurability of $\sqrt{2}$	4
1.4	The Euclidean Algorithm	4
1.5	The Golden Ratio	5

CHAPTER 2

PRIMES AND PERFECT NUMBERS PAGE 7

2.1	The Indivisible Numbers	7
2.2	Perfect Numbers	8
2.3	The Fundamental Theorem of Arithmetic	8
2.4	Mersenne Primes and Euler's Theorem	9

Commensurability and the Diagonal

§ 1.1 Magnitudes and Their Measures

Definition 1.1.1: Commensurable Magnitudes

Two magnitudes a and b are **commensurable** if there exists a magnitude u such that $a = mu$ and $b = nu$ for some positive integers m and n ; the magnitude u is called a common measure. If no common measure exists the magnitudes are **incommensurable**.

Example 1.1.1 (The Meno Square)

A square of side 1 has area 1. Socrates asks Meno's slave: which square has exactly double the area? A side of length 2 gives area 4 — too large. A side of $\frac{3}{2}$ gives $\frac{9}{4}$ — still wrong. The slave is perplexed. Socrates then draws the diagonal of the original square and shows that four right triangles, each half the original square, fit exactly into the square built on that diagonal. The desired side is the diagonal itself — a magnitude of length $\sqrt{2}$. As we shall prove, it is incommensurable with the unit side.

Note:-

The Meno demonstration does not merely solve a geometry problem. Socrates takes it as evidence of *anamnesis*: the slave, who was never taught geometry, recollects the truth through questioning alone. Whether the same soul carries mathematical knowledge across lives is a separate inquiry — but the geometry itself admits no such ambiguity.

§ 1.2 The Pythagorean Theorem

Theorem 1.2.1 Pythagorean Theorem

In a right triangle with legs a, b and hypotenuse c ,

$$a^2 + b^2 = c^2.$$

Proof: Arrange four congruent copies of the right triangle inside a square of side $a + b$, with hypotenuses forming an inner tilted square of side c . Equating areas:

$$(a + b)^2 = 4 \cdot \frac{1}{2}ab + c^2 \implies a^2 + 2ab + b^2 = 2ab + c^2 \implies a^2 + b^2 = c^2.$$

■

Corollary 1.2.2 Length of the Diagonal

The diagonal of a unit square has length $\sqrt{2}$.

Proof: Apply Theorem 1.2.1 with $a = b = 1$: $c^2 = 2$, so $c = \sqrt{2}$. ■

§ 1.3 The Incommensurability of $\sqrt{2}$

Lemma 1.3.1 Parity of Squares

For any integer n : n^2 is even if and only if n is even.

Proof: ($\Leftarrow$) If $n = 2k$ then $n^2 = 4k^2$, which is even.

($\Rightarrow$) We prove the contrapositive. If n is odd, write $n = 2k + 1$. Then $n^2 = 4k^2 + 4k + 1$, which is odd. ■

Theorem 1.3.2 The Diagonal is Incommensurable with the Side

$\sqrt{2}$ is irrational; the diagonal of a unit square is incommensurable with its side.

Proof: Suppose $\sqrt{2} = p/q$ with $p, q \in \mathbb{Z}^+$ and $\gcd(p, q) = 1$. Squaring gives $p^2 = 2q^2$, so p^2 is even. By Lemma 1.3.1, p is even; write $p = 2r$. Substituting: $4r^2 = 2q^2$, hence $q^2 = 2r^2$, so q^2 is even, and again by Lemma 1.3.1, q is even. Then $2 \mid \gcd(p, q)$, contradicting $\gcd(p, q) = 1$. ■

Alternate Proof: Suppose the ratio $p : q$ (in lowest terms) satisfies $p^2 = 2q^2$. From the square of side p , cut off a square of side q in each corner. The remaining L-shaped region has the same ratio $p : q$ as the original. But its dimensions are $p - q$ and $2q - p$, which are strictly smaller positive integers with $(p - q)^2 = 2(2q - p)^2$. This descent never terminates — a contradiction. ■

Wrong Concept 1.1: The Pythagorean Dogma

The early Pythagoreans held that *all is number* — every ratio of magnitudes is a ratio of whole numbers. Theorem 1.3.2 shattered this doctrine. Tradition records that Hippasus of Metapontum perished at sea for disclosing the existence of incommensurables. One should not confuse the beauty of a belief with its truth: the cosmos contains magnitudes that no fraction can name.

Question 1.1: How Close Can Rationals Get?

We have shown $\sqrt{2}$ is irrational. Yet rational numbers are dense in $\mathbb{R}$: for any $\varepsilon > 0$, does there exist $p/q \in \mathbb{Q}$ with $|\sqrt{2} - p/q| < \varepsilon$? Can you find such a fraction for $\varepsilon = 1/100$? For $\varepsilon = 1/1000$?

§ 1.4 The Euclidean Algorithm

Definition 1.4.1: Greatest Common Divisor

The **greatest common divisor** $\gcd(a, b)$ of two positive integers a and b is the largest positive integer dividing both. Geometrically, $\gcd(a, b)$ is the length of the longest rod that measures both a and b without remainder — the greatest common measure of the two lengths.

Example 1.4.1 (A Computation)

To find $\gcd(48, 18)$: divide repeatedly, keeping the remainder.

$$48 = 2 \cdot 18 + 12, \quad 18 = 1 \cdot 12 + 6, \quad 12 = 2 \cdot 6 + 0.$$

The last nonzero remainder is 6, so $\gcd(48, 18) = 6$.

Theorem 1.4.1 The Euclidean Algorithm

Given positive integers $a \geq b$, repeatedly replace (a, b) by $(b, a \bmod b)$ until the remainder is zero. The last nonzero remainder equals $\gcd(a, b)$. The algorithm terminates after at most $2 \log_{\phi} a$ steps.

Sketch of Proof: Each remainder $r_k = r_{k-2} - q_k r_{k-1}$ satisfies $r_k < r_{k-1}$, so the sequence of remainders is strictly decreasing and reaches zero. At each step $\gcd(r_{k-2}, r_{k-1}) = \gcd(r_{k-1}, r_k)$, so the gcd is preserved throughout. The bound on the number of steps follows from the fact that two consecutive remainders decrease by at least a factor of ϕ . ■

Theorem 1.4.2 Bézout's Identity

For any positive integers a and b , there exist integers s and t (possibly negative) such that

$$\gcd(a, b) = sa + tb.$$

Proof: Let d be the smallest positive element of the set $S = \{xa + yb : x, y \in \mathbb{Z}\}$. Dividing a by d with remainder: $a = qd + r$ with $0 \leq r < d$. Then $r = a - qd \in S$, and by minimality of d we have $r = 0$, so $d \mid a$. Likewise $d \mid b$. Hence d is a common divisor. Any common divisor of a and b also divides $sa + tb = d$, so $d = \gcd(a, b)$. ■

Note:-

Euclid described the algorithm as *anthypharesis* — mutual subtraction. Instead of dividing, one repeatedly removes the smaller magnitude from the larger until one magnitude measures the other exactly. The Greeks framed this as a geometric process on line segments rather than arithmetic on integers; the key insight is the same.

§ 1.5 The Golden Ratio

Definition 1.5.1: The Golden Ratio

A line segment is divided in the **golden ratio** if the ratio of the whole to the larger part equals the ratio of the larger part to the smaller. Setting the larger part to 1 and the whole to ϕ , this condition reads $\phi/1 = 1/(\phi - 1)$, giving

$$\phi^2 = \phi + 1, \quad \phi = \frac{1 + \sqrt{5}}{2} \approx 1.618 \dots$$

Example 1.5.1 (The Diagonal of a Regular Pentagon)

In a regular pentagon with side length 1, each diagonal has length ϕ . Moreover, any diagonal intersects two others, cutting each in the golden ratio — and the inner pentagon so formed is again regular, with diagonals of length ϕ^{-1} . This self-similar structure captured the Pythagoreans, who used the pentagram as their secret symbol.

Theorem 1.5.1 The Golden Ratio is Irrational

$\phi = \frac{1 + \sqrt{5}}{2}$ is irrational.

Proof: Since $\phi^2 = \phi + 1$, if $\phi = p/q$ in lowest terms then $p^2/q^2 = p/q + 1 = (p+q)/q$, giving $p^2 = q(p+q)$. Thus $q \mid p^2$. Since $\gcd(p, q) = 1$ we get $q \mid 1$, so $q = 1$ and $\phi = p \in \mathbb{Z}$. But $1 < \phi < 2$, a contradiction. ■

Lemma 1.5.2 The Continued Fraction of ϕ

$\phi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{\ddots}}}$; that is, $\phi = [1; 1, 1, 1, \dots]$ in continued fraction notation.

Proof: From $\phi = 1 + 1/\phi$ (which rearranges $\phi^2 = \phi + 1$), iterating the substitution $\phi \mapsto 1 + 1/\phi$ gives $\phi = 1 + 1/(1 + 1/(1 + \dots))$. The convergents are ratios of consecutive Fibonacci numbers: $1/1, 2/1, 3/2, 5/3, 8/5, 13/8, \dots$ ■

Corollary 1.5.3 Worst-Case Rational Approximation

Among all irrationals, ϕ is the hardest to approximate by rationals: for every rational p/q ,

$$\left| \phi - \frac{p}{q} \right| > \frac{1}{\sqrt{5} q^2}.$$

No irrational has a worse approximation constant than $1/\sqrt{5}$.

Note:-

Plato's *Timaeus* constructs the cosmos from two types of right triangle. One of them, with sides $1, \sqrt{3}, 2$, tiles to form the equilateral triangle — the face of the tetrahedron, octahedron, and icosahedron. The icosahedron's construction relies on the regular pentagon, and hence on ϕ . Whether Plato was aware of the incommensurability of ϕ is unclear; the connection to [Theorem 1.5.1](#) would not have pleased him.

Primes and Perfect Numbers

§ 2.1 The Indivisible Numbers

Definition 2.1.1: Prime Number

An integer $p \geq 2$ is **prime** if its only positive divisors are 1 and p itself. An integer $n \geq 2$ that is not prime is **composite**. By convention, 1 is neither prime nor composite.

Example 2.1.1 (The First Primes)

The primes begin: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, ... The number 2 is the unique even prime; all larger primes are odd. As the numbers grow, primes appear to thin out — which raises the question Euclid answered definitively.

Claim 2.1.1 Every Integer Has a Prime Divisor

Every integer $n \geq 2$ has at least one prime divisor.

Proof: Let p be the smallest divisor of n greater than 1. Were p composite, it would possess a divisor d with $1 < d < p$, and $d \mid p \mid n$ would contradict the minimality of p . Hence p is prime. ■

Theorem 2.1.1 Infinitude of Primes

There are infinitely many prime numbers.

Proof: Suppose for contradiction that $p_1, p_2, \dots, p_n$ exhausts all primes. Set $N = p_1 p_2 \cdots p_n + 1$. For each i , the prime p_i divides $p_1 \cdots p_n$ and hence does not divide N . Yet $N \geq 2$, so by Claim 2.1.1 N has a prime factor — one absent from our list. Contradiction. ■

Proof Idea: The number N above need not itself be prime. For instance, $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031 = 59 \times 509$. The argument only requires that N carries a prime factor *new* to the list — which it always does. ■

§ 2.2 Perfect Numbers

Definition 2.2.1: Perfect Number

A positive integer n is **perfect** if it equals the sum of its proper divisors. Equivalently, writing $\sigma(n)$ for the sum of all positive divisors of n : n is perfect iff $\sigma(n) = 2n$.

Example 2.2.1 (The First Perfect Numbers)

- $6 = 1 + 2 + 3$, with divisors 1, 2, 3, 6.
- $28 = 1 + 2 + 4 + 7 + 14$, with divisors 1, 2, 4, 7, 14, 28.
- 496 and 8128 follow next.

Nicomachus of Gerasa called perfect numbers “*neither excessive nor deficient, but hitting the mean between extremes*” — language Socrates would recognise from his account of virtue.

Theorem 2.2.1 Euclid’s Construction of Even Perfect Numbers

If $q = 2^p - 1$ is prime, then $n = 2^{p-1}(2^p - 1)$ is perfect.

Proof: Since q is an odd prime, $\gcd(2^{p-1}, q) = 1$ and σ is multiplicative, so

$$\sigma(n) = \sigma(2^{p-1}) \cdot \sigma(q) = (1 + 2 + \cdots + 2^{p-1})(1 + q) = (2^p - 1) \cdot 2^p = 2n.$$

Hence n is perfect. ■

Open Question 2.2.1: The Question of Odd Perfect Numbers

Does there exist an odd perfect number?

This question has stood open since antiquity. Euler proved any odd perfect number must take the form $n = q^{4k+1}m^2$ where $q \equiv 1 \pmod{4}$ is prime with $q \nmid m$. We now know any odd perfect number must exceed 10^{1500} , have at least 101 prime factors counting multiplicity, and possess no fewer than 10 distinct prime factors. None has ever been found.

Note:-

In the *Laws*, Plato proposes $5040 = 7!$ as the ideal number of households in a city-state, praising it for having 60 divisors — more than any smaller number. $\sigma(5040) = 14\,400 \neq 2 \times 5040$, so 5040 falls short of perfection. Whether Plato noticed is not recorded; one suspects Socrates would have asked.

§ 2.3 The Fundamental Theorem of Arithmetic

Theorem 2.3.1 Euclid’s Lemma

If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof: Suppose $p \nmid a$. Since p is prime, $\gcd(p, a) = 1$. By Theorem 1.4.2, there exist integers s, t with $sp + ta = 1$. Multiplying by b : $spb + tab = b$. Now $p \mid spb$ and $p \mid tab$ (since $p \mid ab$), so $p \mid b$. ■

Corollary 2.3.2 Extension to Prime Powers

If p is prime and $p \mid a_1 a_2 \cdots a_k$, then $p \mid a_i$ for some $i \in \{1, \dots, k\}$.

Proof: Apply [Theorem 2.3.1](#) inductively: either $p \mid a_1$ or $p \mid a_2 \cdots a_k$. Repeating on the second factor terminates in at most k steps. ■

Theorem 2.3.3 Fundamental Theorem of Arithmetic

Every integer $n \geq 2$ can be written as a product of primes, and this factorisation is unique up to the order of factors.

Proof: *Existence.* By induction: if n is prime we are done. Otherwise $n = ab$ with $1 < a, b < n$, and by induction each of a, b factors into primes.

Uniqueness. Suppose $p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$. Since p_1 divides the left side it divides the right, so by [Corollary 2.3.2](#) $p_1 \mid q_j$ for some j . As q_j is prime, $p_1 = q_j$. Cancel p_1 from both sides and repeat. This forces $r = s$ and the p_i to be a permutation of the q_j . ■

Corollary 2.3.4 Square Roots of Primes are Irrational

For every prime p , $\sqrt{p}$ is irrational.

Proof: Suppose $\sqrt{p} = a/b$ in lowest terms. Then $pb^2 = a^2$. By [Theorem 2.3.3](#), the prime p appears to an odd power on the left (it divides b^2 an even number of times and contributes one extra factor) but to an even power on the right. This contradicts uniqueness of factorisation. ■

Wrong Concept 2.1: All Integers Factor, But Not Always Uniquely

In more general algebraic systems, unique factorisation can fail. In $\mathbb{Z}[\sqrt{-5}]$, for instance, $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, and all four factors are irreducible. [Theorem 2.3.3](#) is special to $\mathbb{Z}$, and it depends critically on [Theorem 2.3.1](#) — which itself rests on [Theorem 1.4.2](#). Remove any link and the chain of uniqueness breaks.

§ 2.4 Mersenne Primes and Euler's Theorem

Definition 2.4.1: Mersenne Prime

A **Mersenne prime** is a prime of the form $M_p = 2^p - 1$. The exponent p must itself be prime, as the following result shows.

Theorem 2.4.1 Necessary Condition for Mersenne Primes

If $2^n - 1$ is prime, then n is prime.

Proof: Suppose $n = ab$ with $1 < a, b < n$. Using the factorisation $x^a - 1 = (x - 1)(x^{a-1} + \cdots + 1)$ with $x = 2^b$:

$$2^n - 1 = (2^b)^a - 1 = (2^b - 1)((2^b)^{a-1} + \cdots + 1).$$

Both factors exceed 1, so $2^n - 1$ is composite. Contrapositive gives the result. ■

Example 2.4.1 (Small Mersenne Primes and Composites)

- $M_2 = 3$, $M_3 = 7$, $M_5 = 31$, $M_7 = 127$ are prime.
- $M_{11} = 2047 = 23 \times 89$ is composite despite 11 being prime. [Theorem 2.4.1](#) gives a necessary condition, not sufficient.
- The largest known prime (as of 2024) is $M_{136\,279\,841}$, a number with over forty million decimal digits.

Theorem 2.4.2 Euler's Characterisation of Even Perfect Numbers

Every even perfect number has the form $2^{p-1}(2^p - 1)$ where $2^p - 1$ is prime.

Sketch of Proof: Let $n = 2^k m$ be even and perfect with m odd and $k \geq 1$. Since σ is multiplicative and $\sigma(n) = 2n$,

$$\sigma(2^k) \sigma(m) = 2^{k+1} m, \quad \text{i.e.,} \quad (2^{k+1} - 1) \sigma(m) = 2^{k+1} m.$$

Set $q = 2^{k+1} - 1$ (odd). Then $q \mid m$; write $m = qj$. Substituting: $\sigma(m) = 2^{k+1} j$. If $j > 1$, then $1, j, q, qj$ are four distinct divisors of m , giving $\sigma(m) \geq 1 + j + q + qj = (1 + j)(1 + q) > 2^{k+1} j$, a contradiction. Hence $j = 1$, $m = q$, and $\sigma(m) = m + 1 = 2^{k+1}$, forcing $q = 2^{k+1} - 1$ to be prime. ■

Open Question 2.4.1: Are There Infinitely Many Mersenne Primes?

It is unknown whether there are infinitely many Mersenne primes, and hence infinitely many even perfect numbers. Euclid's and Euler's theorems together give a complete characterisation of even perfect numbers — every one arises from a Mersenne prime — but the supply of Mersenne primes remains mysterious. The distributed computing project GIMPS has found the largest known primes for decades; as of 2024, only 51 Mersenne primes are known.

Note:-

The correspondence between Mersenne primes and even perfect numbers, established jointly by Euclid ([Theorem 2.2.1](#)) and Euler ([Theorem 2.4.2](#)), spans two millennia. Euclid gave the construction; Euler proved there is nothing else. That a question posed in Alexandria around 300 BC was only fully answered in 1772 AD speaks to the patience mathematics demands — a virtue Socrates would have approved of.